Lösung: Rollende Vollkugel mit Rollreibung

a) Energieerhaltung

$$E_{FP} = E_H + \Delta Q$$

(E_{FP}: Energie am Hochpunkt; E_H: Energie am Fußpunkt; ΔQ: Wärmeenergie)

mit $E_{FP} = \frac{1}{2} M v_0^2 + \frac{1}{2} J_K \cdot \omega_0^2$

$= \frac{1}{2} M v_0^2 + \frac{1}{2} J (v_0/R)^2 = \frac{1}{2} (M + J/R^2) v_0^2$

mit $J = \frac{2}{5} M R^2$

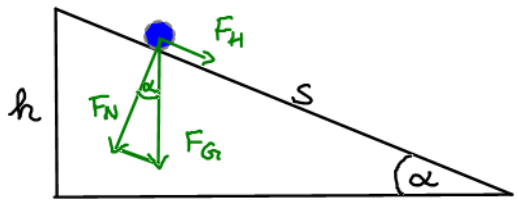
$$E_{FP} = \frac{1}{2} (M + \frac{2}{5} M) v_0^2 = \underline{\underline{\frac{7}{10} M v_0^2}}$$

$$E_H = M \cdot g \cdot h$$

$$\Delta Q = -\Delta W = -\vec{F}_R \cdot \vec{s} = F_R \cdot s = \mu \cdot F_N \cdot s$$

$$= \mu (\cos(\alpha) F_G) \frac{h}{\sin(\alpha)}$$

$$= \frac{\mu M \cdot g \cdot h}{\tan(\alpha)}$$



$\sin(\alpha) = \frac{h}{s}$ & $\cos(\alpha) = \frac{F_N}{F_G}$

$$\Leftrightarrow \quad \frac{7}{10} \cancel{M} v_0^2 = \cancel{M} \cdot g \cdot h + \frac{\mu \cancel{M} \cdot g \cdot h}{\tan(\alpha)}$$

$$\frac{7}{10} v_0^2 = g \cdot h \left(1 + \frac{\mu}{\tan(\alpha)}\right)$$

$$\underline{\underline{h = \frac{7}{10} \frac{v_0^2}{g \cdot (1 + \mu/\tan(\alpha))}}}$$

$$h \approx \frac{7}{10} \cdot \frac{(5{,}173\,m/s)^2}{9{,}81\,m/s^2 \cdot (1 + 0{,}022/\tan(5°))} \approx \underline{\underline{1{,}87\,m}}$$

b)

$$P = \frac{\Delta Q}{E_{FP}} = \frac{\frac{\mu M \cdot g \cdot h}{\tan(\alpha)}}{\frac{7}{10} M v_0^2} = \frac{10}{7} \frac{\mu g \cdot h}{v_0^2 \cdot \tan(\alpha)}$$

$$= \frac{10}{7} \frac{\mu g}{v_0^2 \cdot \tan(\alpha)} \frac{7}{10} \frac{v_0^2}{g \cdot (1 + \mu / \tan(\alpha))} = \frac{\mu}{\tan(\alpha) + \mu}$$

$$P \approx \frac{0{,}022}{\tan(5°) + 0{,}022} \approx 0{,}201 \approx \underline{\underline{20{,}1\,\%}}$$